

,Princeton University
Mathematical Intermediate Microeconomics (ECO 310)
Midterm Spring 2024

Instructions: You have 85 minutes to complete this exam, plus 10 minutes to upload your answers. There are 10 multiple choice questions and 2 short answer questions. **In the Multiple Choice section, simply indicate the correct answer. In the Short Answer section, show your work and explain your answers for full credit!**

- The exam is open-note, and calculators are permitted. However, please work **alone**.
- **If you are unsure whether you're interpreting a question correctly, state your interpretation and proceed (or email me if you prefer).**

The honor code applies, as usual.

Q1: Multiple Choice (40 points, 4 each)

1. Jake cares about three attributes of cars: price, quality, and color. When comparing two cars, he prefers whichever is better in two attributes. Jake is:
 - (a) Rational
 - (b) Irrational because these preferences are incomplete
 - (c) Irrational because these preferences are intransitive
 - (d) Two of these are correct
2. In which of the following situations *might* (depending on prices and income) it be optimal to spend your entire budget on good 1?
 - (a) If $u(x_1, x_2) = x_1 + x_2^2$
 - (b) If $u(x_1, x_2) = x_1^2 + x_2$
 - (c) If $u(x_1, x_2) = x_1^2 x_2$
 - (d) In two of these situations
 - (e) In three of these situations
3. A consumer's preferences over good 1 and good 2 are described by $u(x_1, x_2) = x_1 x_2^3$. If this consumer behaves optimally, he will:
 - (a) Buy 3 times as much good 2 as good 1.
 - (b) Buy 4 times as much good 2 as good 1.
 - (c) Spend 4 times as much on good 2 as good 1.
 - (d) None of these is correct.
4. A consumer initially buys 5 oranges and 10 apples. Then an apple tax increases the price per apple by \$0.50, but the consumer is at the same time given a \$5 fruit subsidy. Which of the following best describes this situation?
 - (a) The consumer is most likely better off now (with the tax and subsidy) than he was originally.
 - (b) The consumer is most likely worse off now (with the tax and subsidy) than he was originally.
 - (c) The consumer is definitely indifferent between his original and current situations.
5. A consumer gets utility $u(L, R) = \min\{L, R\}$ from L left shoes and R right shoes. If $m = 10$, $p_L = 1$, and p_R increases from 1 to 4, the change in demand for right shoes due to the income effect is:
 - (a) 0
 - (b) -1
 - (c) -2
 - (d) -3

- (e) None of these is correct
6. A consumer expects to end up with final wealth $w_0 - 200$ if an accident occurs (this happens with probability p), and $w_0 + 100$ otherwise. They tell you that their expected utility from accepting this gamble is 100. Which of the following would let you conclude that the consumer is risk-averse?
- (a) If you knew that $u(w_0 + 100 - 300p) > 100$
 - (b) If you knew that $u(w_0 + 100 - 300p) < 100$
 - (c) If you knew that $u(w_0 - 200) < 100$
 - (d) None of these is correct
7. In Lecture 10, we saw the following “paradox”: many people prefer saving 200 of 600 people for sure, to a risky option with a $\frac{1}{3}$ chance of saving all 600 people and a $\frac{2}{3}$ chance of saving nobody. But they prefer a risky option with a $\frac{1}{3}$ chance of killing nobody and a $\frac{2}{3}$ chance of killing everybody, to an option that kills 400 people for sure.
- If people care about the # people alive in the end, then
- (a) This is inconsistent with EU maximization
 - (b) This violates consequentialism
 - (c) This violates independence
 - (d) Two of these are true
 - (e) All of these are true
8. Which of the following would a risk-averse consumer **definitely** prefer to a 50-50 \$0 or \$200 gamble? Pick **ALL** that apply.
- (a) \$90 lump-sum
 - (b) \$110 lump-sum
 - (c) A 50-50 \$50 or \$150 gamble
 - (d) A 50-50 \$10 or \$190 gamble.
9. On Monday, Homer Simpson buys a bundle of food and drink that costs \$100. On Tuesday, he buys a different bundle, for \$80, which *would have cost \$100 on Monday*. In which of the following situations does Homer violate WARP?
- (a) If the Monday bundle would cost more than \$80 at Tuesday’s prices.
 - (b) If the Monday bundle would cost less than \$80 at Tuesday’s prices.
 - (c) If the Monday bundle would cost less than \$100 at Tuesday’s prices
 - (d) Homer does not violate WARP
 - (e) Not enough information given.
10. Jessica has perfect substitute preferences over pens and pencils: she wants as many things to write with as possible, but doesn’t care whether pens or pencils. She sets up a Lagrangian to find her best affordable bundle. Which of the following best describes which non-negativity constraints will bind, assuming **pens are cheaper**?

- (a) Non-negativity constraints bind for both pens and pencils
- (b) Non-negativity constraints are NON-binding for both pens and pencils.
- (c) Non-negativity constraints bind for pencils but not pens
- (d) Non-negativity constraints bind for pens but not pencils
- (e) Not enough information given to answer this question

Short Answer

Q2: Expected Utility

Consider a decision-maker with utility from wealth $u(w) = w^{1-\alpha}$.

- a. (10) For a gamble that offers equal chances of either \$0 or \$200, write out the equation describing the certainty equivalent (assuming no initial wealth for this part only). If α increases, does the certainty equivalent increase or decrease?
- b. Now suppose the DM can invest as much of their initial wealth w_0 as they like in a double-or-nothing asset: their investment doubles in value with chance p , but if this doesn't happen, they lose everything they invested. (For example, \$100 either grows to \$200 or vanishes).
 - (a) (8) Write out the expected utility from investing \$x.
 - (b) (12) Find (or characterize) the optimal investment \$x.

Q3: Demand

Melsi (my puppy) has preferences over cookie consumption today (C_1) and cookie consumption tomorrow (C_2) given by $u(C_1, C_2) = v(C_1) + v(C_2)$, where $v : \mathbb{R} \rightarrow \mathbb{R}$ is an *increasing and concave* function (eg log or square root).

She has w_1 cookies today. She expects any cookies that she saves to magically double overnight, but otherwise does not expect to receive any additional cookies tomorrow.

- a. (5) Write out her budget constraint. (If she eats C_1 cookies today, how many does she save, and then how many can she eat tomorrow?)
- b. (6) Are her preferences over bundles (C_1, C_2) convex? Explain.
- c. (9) Describe, as precisely as you can, how to find her best affordable bundle (C_1, C_2) as a function of w_1 . (You can't solve for it exactly without knowing v , but write out the equations you would solve if you did know v . You may use any method).
- d. (10) For this part only, specialize to $v(C) = \ln C$. What bundle (C_1, C_2) most efficiently i.e. cheaply attains utility u^* ?